Transscleral Resection versus Iodine Brachytherapy for Choroidal Malignant Melanomas 6 Millimeters or More in Thickness

A Matched Case–Control Study

Tero Kivelä, MD,[1] Ilkka Puusaari, MD,[1] Bertil Damato, MD[2]

Objective: To assess differences in treatment outcome after transscleral resection (TSR) and iodine brachytherapy (IBT) of choroidal melanomas too thick for ruthenium brachytherapy.

Design: Matched case–control study.

Setting: Two national ocular oncology services.

Participants: Forty-nine pairs of patients with choroidal melanoma 6 mm thick or more (range = 6–14), treated within 3 years of each other by either TSR or IBT, and retrospectively matched by age, visual acuity at diagnosis, largest basal tumor diameter, and posterior extension of the tumor.

Intervention: Iodine brachytherapy in Finland or TSR in United Kingdom. Prospectively collected time-to-event data were compared with parametric Weibull regression, adjusting for matching.

Main Outcome Measures: Tumor control, complications, and visual acuity.

Results: Risk of local recurrence after IBT was smaller than that after TSR (hazard ratio [HR] = 0.02, 95% confidence interval [CI] = 0.01–0.11, $P<0.001$, Weibull regression adjusted for matching), but the 8-year all-cause and melanoma-specific (HR = 0.81, 95% CI = 0.30–2.22, $P = 0.69$) survivals did not differ. The risks of cataract (HR = 2.05, 95% CI = 1.08–3.89, $P = 0.029$), maculopathy (HR = 2.28, 95% CI = 0.96–5.43, $P = 0.062$), and vitreous hemorrhage (HR = 2.30, 95% CI = 0.95–5.57, $P = 0.064$) were higher after IBT. Rubeosis, neovascular glaucoma, and optic neuropathy developed only after IBT. Risk of retinal detachment, exudative after IBT and rhegmatogenous after TSR (HR = 0.84, 95% CI = 0.40–1.75, $P = 0.63$), and risk of losing 20/60 vision (HR = 1.37, 95% CI 0.76–2.45, $P = 0.29$) were comparable between the groups, but risk of losing 20/200 vision was higher after IBT (HR = 2.38, 95% CI = 1.48–3.83, $P<0.001$). No overall difference in quality of life was found.

Conclusions: This study suggests that TSR preserves 20/200 vision better than IBT and avoids some of its major complications, but increases the risk of local recurrence. About 350 patients would need to be randomized to prove that TRS preserves 20/60 vision better than IBT. *Ophthalmology 2003;110:2235–2244 © 2003 by the American Academy of Ophthalmology.*

There is no consensus on the best treatment of choroidal melanomas with a thickness of 6 mm or more, except that ruthenium plaque brachytherapy (RuBT) is generally considered inadequate.[1–3] The therapeutic options include iodine[4–9] and palladium plaque brachytherapy,[10] charged particle teletherapy, stereotactic radiotherapy, transscleral local resection,[11,12] and enucleation. The published literature comparing any 2 of these treatments is scanty,[5,9,13–17] and thus there is a need for further controlled studies.

The best method of comparing 2 forms of treatment for

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[1] Oncology and Vitreoretinal Service, Department of Ophthalmology, Helsinki University Central Hospital, Helsinki, Finland.

[2] Ocular Oncology Service, St. Paul's Eye Unit, Royal Liverpool University Hospital, Liverpool, United Kingdom.

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Correspondence to Bertil Damato, MD, Ocular Oncology Service, St. Paul's Eye Unit, Royal Liverpool University Hospital, Prescot Street, Liverpool L7 8XP, United Kingdom. E-mail: Bertil@Damato.co.uk.

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